Declaring Variables

When setting up a new variable, we need to at the very least determine two pieces of information: the variables data type and its name. This process is called **declaring** a variable where we will hold a place open in the Arduino's memory to store our information. Declaring a variable is as simple as indicating the data type followed by a unique name for the variable. The following are some examples of simple variable declarations:

```
int x;

int myVariable;

int time;
```

The sort of data that can be stored in a variable, and what can be done with it, is determined by the variable's **data type**. The data type that is chosen is very important later on when it affects what we can and cannot do with it. In the previous examples, we are declaring integer data types, signified by the `int` keyword, a useful data type for numerical data. After establishing the data type, we need a clear **variable name** that tells us in plain English what the variable does. As a general rule of thumb, it should be descriptive, but not overly cumbersome. Take the following examples:

```
int r;

int red;

int redLEDConnectedToPin9;
```

Each of these three variable names is technically correct, but naming our variable `red` instead of the letter `r` is generally easier to understand. Single-letter variable names, with a few exceptions, are often harder to read and make sense of; for example, a lowercase `l` (L) looks an awful lot like the uppercase `I` (I), or even the number `1` (1). The extreme case, `redLEDConnectedToPin9`, is generally too verbose and makes reading, as well as writing, the code somewhat tedious.

Where these previous examples simply declare the variable data type and variable name, it is also possible to assign a value to a variable when we declare it in a single statement. In our example sketch, we did this in the following line:

```
int red = 9;
```

Here we are declaring the variable named `red` as an integer data type and assigning the value `9` to that variable name using the assignment operator to be discussed shortly. While the Arduino compiler will still hold a place open in its memory for a variable even if we do not assign it a value, it is generally a good idea to assign a value in the variable declaration even if that value is 0.

Variable Names

We can name a variable pretty much anything we want, but there are a few rules we need to follow. First of all, function and variable names can only contain letters, underscores, numbers, or the dollar sign. Spaces or odd characters like the @ or & are not allowed and the first character cannot be a number. White space is not allowed inside a variable name, as generally when white space is allowed, you could have one space or hundreds and it wouldn't matter. This white space includes new lines from hitting the Return or Enter keys, or indents from the space bar or Tab key. Considering this, the variable names